

What decreases the TFP ? The aging labor and ICT imbalance

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Abstract

As empirically found by Lebow (1993), Sbordone (1996) and Basu and Fernand (2001), among others, the Total Factor Productivity (TFP) account for a large percent of growth within economies and behaves pro-cyclically in a business cycle. The TFP is often seen as the real driver of growth within an economy. Basu and Fernand (2001) provide four channels from the TFP to the business. They perceive the TFP changes as exogenous shocks for the business cycle. The next logical step is to investigate what causes the TFP changes and why the TFP decreases in some countries. Hayashi and Prescott (2002) found that the low TFP accounts well for the Japanese lost decade of growth, and conjectured that the low productivity growth was the result of a policy that subsidized inefficient firms and declining industries. This policy resulted in lower productivity because the inefficient producers produce a greater share of the output. Though many other papers have mentioned the TFP decrease in the business slump, they have not investigated what decreased the TFP. Exceptions are Jorgenson and Motohashi (2005), Jorgenson and Nomura (2005), and Hayashi and Nomura (2005), who mostly expand a single-sector growth model of Hayashi and Prescott to a multi-sector model, focusing on the relationship between information technology, labor input and the TFP through the price mechanism. They did not include the idiosyncratic characters like aging and skill levels directly into the physical inputs of labor and capital, and did not estimate the effect of each factor on the TFP. Based on the natural selection mechanism of economic Darwinism, Nishimura et.al. (2005) found that efficient firms in terms of TFP exited while inefficient ones survived. Further, this phenomenon is observed mainly for new entrants and contributes substantially to a fall in macro TFP after 1996. The work of Fukao, et.al (2005) is in the same strand.

The purpose of this paper is to investigate what decreases the TFP and why the TFP decrease in some countries, focusing on the quality of labor and capital inputs. We do this by using cross country data for the manufacturing industries of some OECD countries.

Our framework includes the aging of the physical labor input and the information technology component of the physical capital input using a framework related to Hellerstein et al. (1999) and Ilmakunnas and Maliranta (2005). We then can identify the effect of each factor of labor characteristics on the TFP, while we have no growth theory basis used by the previous researches. We also provide a comprehensive investigation based on two hypotheses of the relationship between workers of different ages, substitutability and complementarity, by using cross country data. This is different from

different from the literature on worker characteristics and productivity, where labor substitutability is assumed and the analysis is conducted with single country data.

We find that (i) Based on the substitutability hypothesis among age groups, the low TFP is related to the aging of the labor input and to insufficient information technology content in the capital input. The aging of the labor input improves the productivity only in the low skilled labor category, but contributes negatively to productivity in the medium to high skilled labor category. Due to these positive and negative effects of aging and ICT imbalance, the TFP has since 1990 increased in Austria, Finland, Netherlands, and USA, while it has stagnated or decreased in Australia, Denmark, and Japan. (ii) Based on the complementarity hypothesis among the age groups, the increased share of the old workers has improved the productivity of the young and mid-aged workers in the low skilled labor category, while it has had no effect in the medium to high skilled group. The aging process has had positive effects in the low-skilled labor group in Japan and Finland, consistent with the results based on the substitutability hypothesis.

With these findings we provide a new explanation of a low TFP, which differs from the previous viewpoints suggested by Hayashi and Prescott (2002), Jorgenson and Motohashi (2005) and Hayashi and Nomura (2005). Moreover, we provide an empirical study of why the TFP has stagnated or increased in some country, based on the idiosyncratic feature in aging labor and ICT imbalance.

Keywords: TFP; quality of labor and capital, substituteability and complementarity
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